

Fundamentals of Data and Database Management System

Chapters 1 & 2: Data, DBMS, Architecture & Models

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Outline

Part 1: Data & DBMS Fundamentals

- What is Data?
- Types of Data
- Database Concepts
- DBMS Overview
- Advantages & Disadvantages

Part 2: Architecture & Models

- Two-Tier & Three-Tier Arch.
- Data Flow: External to Internal
- Database Models (Hierarchical, Network, Relational, NoSQL)

"Data is the new oil, but unrefined data is just a liability."

What is Data? (1.1)

Definition

Data are raw, unprocessed facts, figures, or symbols that represent objects, events, or concepts in the real world.

Examples

- "John", 20, "2025-05-01", B+
- 150, 75.5, "Blue", 10-12-2024
- Sensor readings: 23.5°C, 1013 hPa

Data alone has no meaning until it is interpreted and given context → **Information**.

Data vs. Information

Data: 35, 25, 40 **Information:** Average temperature = 33.3°C

Types of Data (1.2)

- **Structured Data:** Highly organized, tabular format (RDBMS).
Example: Excel sheet, SQL tables.
- **Semi-structured Data:** No rigid schema but uses tags/markers.
Example: JSON, XML, HTML.
- **Unstructured Data:** No predefined structure.
Example: Text files, images, videos, audio.

Type	Storage	Complexity
Structured	Relational DB	Low
Semi-structured	NoSQL, object stores	Medium
Unstructured	Data lakes, blob storage	High

What is a Database? (1.3)

Definition

A **database** is a shared, integrated collection of logically related data, designed to meet the information needs of an organization.

Properties:

- Persistent storage
- Multiple views
- Concurrent access
- Data integrity

Real-world examples:

- University registration system
- Bank transaction DB
- Airline reservation system
- E-commerce (Amazon, eBay)

Database $\neq$ DBMS (database is the container; DBMS is the software).

What is a DBMS? (1.4)

Definition

A **Database Management System (DBMS)** is a software system that enables users to define, create, maintain, and control access to databases.

Core functions

- 1 Data definition (DDL)
- 2 Data manipulation (DML)
- 3 Data security and integrity
- 4 Concurrency control
- 5 Backup and recovery
- 6 Performance optimization

Popular DBMS: MySQL, PostgreSQL, Oracle, SQL Server, MongoDB.

Advantages of Database (1/2)

- **Reduced data redundancy:** Data stored once; eliminates duplication.
- **Data consistency:** Updates propagate automatically; avoids anomalies.
- **Data sharing:** Multiple users and applications access same data concurrently.
- **Improved data integrity:** Enforce constraints (e.g., NOT NULL, UNIQUE, FK).

Example

Without DBMS, student address stored in 3 files → update inconsistency.
With DBMS, one address record for each student.

Advantages of Database (2/2)

- **Enhanced security:** User authentication, privileges, access control.
- **Backup and recovery:** Automated backups, transaction logs, point-in-time recovery.
- **Data independence:** Logical and physical changes without rewriting applications.
- **Multiple user views:** Each user sees a tailored subset of the database.
- **Standardization:** Uniform data management, SQL standards.

Organizations using DBMS achieve higher data quality and better decision-making.

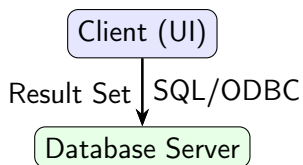
Disadvantages of DBMS

- **High initial cost:** Software licenses, hardware, training.
- **Complexity:** Skilled DBAs and developers required.
- **Performance overhead:** For small, simple applications, a DBMS may be overkill.
- **Increased vulnerability:** Centralization requires robust security; single point of failure (mitigated by replication).
- **Upgrade and migration costs:** Version updates, cloud migration, legacy data conversion.

When to avoid a full DBMS?

Very small applications, read-only static data, embedded systems with limited resources.

Database Architecture: Two-Tier (Client-Server)



Characteristics:

- Client manages presentation & business logic.
- Server handles queries and storage.
- Direct communication (ODBC, JDBC).

Pros: Simple, fast for small user base.

Cons: Scalability issues, heavy clients, security exposure.

Three-Tier Architecture

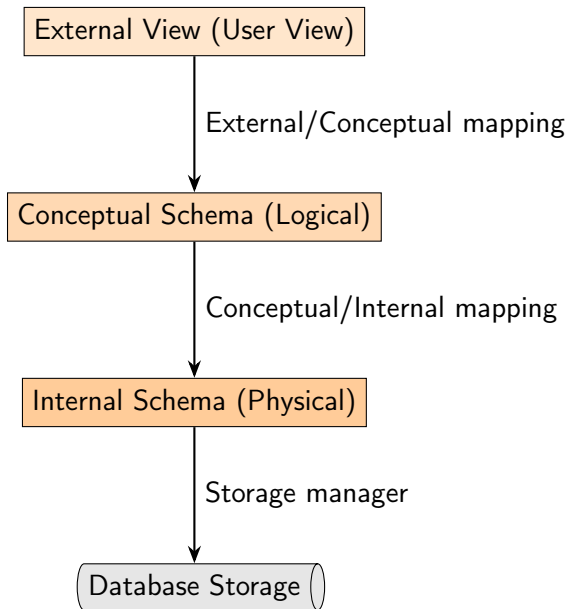


- **Tier 1:** User interface (web/mobile/desktop)
- **Tier 2:** Business logic, processing, validation
- **Tier 3:** DBMS and data storage

Advantages: Scalability, security (no direct DB access), maintainability, reusability.

Examples: Modern web apps, e-commerce, banking portals.

Data Flow: External $\rightarrow$ Internal World



Data flows from high-level abstract views down to physical byte-level

Three-Schema Architecture & Mappings

External Schema Individual user views; hides irrelevant details.

Conceptual Schema Logical structure, entities, relationships, constraints (e.g., ER diagram).

Internal Schema Physical storage: indexes, hashing, data blocks, file organization.

- **External-Conceptual Mapping:** Translates user requests to logical operations.
- **Conceptual-Internal Mapping:** Translates logical records to physical addresses.

Data Independence

Logical independence: Change conceptual schema without altering external views.

Physical independence: Change internal schema without affecting conceptual schema.

Example: Student Query Data Flow

User query (External View):

```
SELECT name FROM student WHERE id = 'S1001';
```

Steps:

- 1 External view: student_view (subset of columns).
- 2 Mapping → Conceptual: STUDENT(id, name, major, ...).
- 3 Conceptual → Internal: INDEX on id, file block #42, offset 128.
- 4 DBMS retrieves physical record and returns name.

— External Storage
— Conceptual
— Internal

Database Models (2.5)

A **database model** defines the logical structure of a database and the relationships among data.

Classic Models:

- Hierarchical
- Network
- Relational (dominant)

Modern/Extended:

- Object-Oriented
- Object-Relational
- NoSQL (document, key-value, graph, column)

Each model solves different data representation needs.

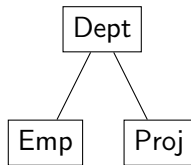
Hierarchical & Network Models

Hierarchical Model (IMS):

- Tree structure with parent-child relationships.
- 1:N relationship; a child has only one parent.
- *Limitation*: Many-to-many relationships require redundancy.

Network Model (CODASYL):

- Graph structure, allows M:N relationships.
- Records linked via pointers (sets).
- *Limitation*: Complex navigation, schema changes affect applications.



Relational Model (Dominant Model)

Definition

Proposed by E.F. Codd (1970). Data represented as **tables (relations)** consisting of rows (tuples) and columns (attributes).

Key properties:

- No duplicate tuples, unordered rows.
- Atomic values (1NF), domains.
- Primary keys, foreign keys enforce integrity.
- Declarative query language: SQL.

StudentID (PK)	Name	Major
101	Alice	CS
102	Bob	IT

Advantages: Simplicity, strong mathematical foundation, data independence.

Object-Oriented & NoSQL Models

Object-Oriented DBMS (OODBMS)

Integrates database capabilities with OOP concepts: objects, classes, inheritance, methods.

Examples: ObjectDB, db4o. Used in CAD, multimedia.

NoSQL Models

For big data, high scalability, flexible schema.

- **Document:** MongoDB (JSON)
- **Key-Value:** Redis, DynamoDB
- **Column-family:** Cassandra
- **Graph:** Neo4j (relationships first)

Key Takeaways

- 1 Data is raw facts; a **database** organizes data; a **DBMS** manages it.
- 2 DBMS advantages: consistency, sharing, security, backup, data independence.
- 3 Database architectures: Two-tier (client-server) and Three-tier (web/app/db).
- 4 Three-schema architecture enables physical and logical data independence via mappings.
- 5 Database models evolved from hierarchical → network → relational → OO/NoSQL.
- 6 The relational model remains most widely used for business applications.

Next topics: Entity-Relationship Modeling, Normalization, SQL.

Questions & Discussion

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References:

- Elmasri & Navathe, “Fundamentals of Database Systems”, 7th Ed.
- Silberschatz, Korth, Sudarshan, “Database System Concepts”, 7th Ed.
- Connolly & Begg, “Database Systems”, 6th Ed.

Thank you!

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